

12.2 The Reverse Process

MSE on the noise you added. T reverse steps. Lab 12 uses the oracle mean.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Generation walks the forward process **backward**. You start from $x_T \sim \mathcal{N}(0, \mathbf{I})$ and, step by step, remove a bit of noise until you have x_0 . By the end you should be able to write the noise-prediction loss, take one reverse-step mean, and say what Lab 12's oracle is replacing.

1. What the diffusion reverse process is

The **reverse** is: predict the noise you added, take a step toward x_0 , repeat T times. A net $\varepsilon_\theta(x_t, t)$ predicts the ε from note **12.1**. Training is ordinary MSE on that noise. Sampling starts from $x_T \sim \mathcal{N}(0, \mathbf{I})$ and denoises.

Ho et al. (DDPM) made noise prediction the default parameterization. Predicting ε is equivalent (up to scaling) to predicting x_0 or the score $\nabla_{x_t} \log q(x_t)$. Pick one parameterization and stay consistent.

You cannot write the true reverse kernel in closed form for images, because it would need the unknown data density. If you **knew** x_0 , the true posterior $q(x_{t-1} | x_t, x_0)$ is Gaussian with a closed-form mean.

Lab 12 uses that **oracle** mean for a single step so you can see the geometry without training a U-Net. A trained ε_θ is what replaces the known ε (and therefore the known x_0) at sample time.

2. Why we use it

MSE on a known ε is easier than a GAN min-max. There is no discriminator to race. The forward jump from note **12.1** already gave you x_t in one line, so the loss never unrolls a reverse chain.

The cost sits at **sample** time. You pay T network calls (or fewer with DDIM / distilled samplers). That is the trade: stable regression training, slow ancestral sampling. GANs (Week 11) sample in one forward pass and fight mode collapse instead.

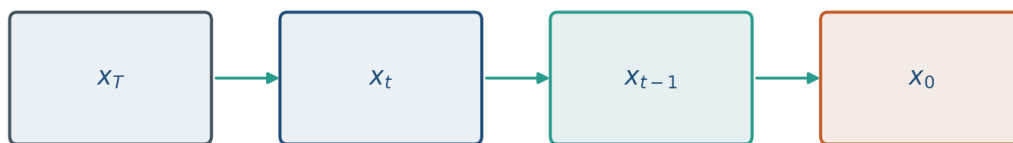
For a project, report the sampler (T , DDPM vs DDIM) next to the metric. Changing T at test time is part of the method. Classifier-free guidance, which steers the same reverse chain with a text condition, is the next note.

3. Architecture

The network is a **denoiser** that sees a noisy input and a timestep: a U-Net or a DiT on x_t with a time embedding. The same x_t is a different problem at $t = 10$ and at $t = 900$, so t is an input, not a footnote.

Lab 12 uses an **oracle** mean so you see the geometry without training. You are not training a U-Net in that lab. You know x_0 , so you can form the true posterior mean and check that one reverse step moved closer.

Reverse: a network predicts the noise (or x_0) and steps toward data



Sampling is an **iterative** procedure. Start at x_T , apply the reverse-step mean (plus a little noise except at $t = 1$), and walk down to x_0 . Each step is one network call once ε_θ exists.

4. How it works, step by step

Training is ordinary regression.

1. Sample x_0 from the data, t uniformly from $1 \dots T$, and $\varepsilon \sim \mathcal{N}(0, \mathbf{I})$.
2. Form x_t with the one-liner from note **12.1**.
3. Predict $\varepsilon_\theta(x_t, t)$. Take MSE against the true ε . No adversary. No reverse chain inside the loss.

Sampling is the reverse walk.

1. Draw $x_T \sim \mathcal{N}(0, \mathbf{I})$.
2. For $t = T, \dots, 1$, plug $\varepsilon_\theta(x_t, t)$ into the reverse-step mean below, and add $\sigma_t z$ for $t > 1$.
3. Return x_0 .

If you knew x_0 , Lab 12 skips the net and uses the true posterior mean for a single step. The lab check is $\|x_t - x_0\|$ versus $\|\hat{x} - x_0\|$ at $t = 20$: the second should be smaller. If $\varepsilon_\theta = 0$ (the net refuses to denoise), the subtracted term vanishes and you mostly rescale x_t , which is not a sample from the data.

5. Mathematical formulas

The training loss is

$$\mathbb{E}_{t, x_0, \varepsilon} \|\varepsilon - \varepsilon_\theta(x_t, t)\|^2.$$

Given x_t and a noise prediction, a typical mean for x_{t-1} is

$$x_{t-1} = \frac{1}{\sqrt{\alpha_t}} \left(x_t - \frac{1 - \alpha_t}{\sqrt{1 - \tilde{\alpha}_t}} \varepsilon_\theta(x_t, t) \right) + \sigma_t z,$$

with $z \sim \mathcal{N}(0, \mathbf{I})$ for $t > 1$, and σ_t from the schedule (often $\sigma_t^2 = \beta_t$). At a generic t you do not jump all the way to x_0 ; you move **closer**. Forgetting $\sigma_t z$ on the way down makes every sample look identical.

6. Positive points and negative points

Positive.

- MSE on ε is ordinary regression, which is easier to stabilize than a GAN min-max.
- Predicting ε is enough: it is equivalent (under the usual parameterization) to predicting the posterior mean.
- Lab 12's oracle mean lets you see one reverse step without training a U-Net.
- You can change T (or switch to DDIM) at sample time and treat the sampler as part of the method.

Negative.

- Ancestral sampling costs many network calls; quality versus speed is a real knob.
- Omitting t as an input makes the same x_t look like one problem at every noise level.
- Dropping $\sigma_t z$ collapses diversity: every run hugs the mean.
- Classifier-free guidance, which steers this chain with text, is extra machinery (next note).

When not to. If you need one-pass samples and can live with collapse, that is still a GAN. If you need text control, do not stop at this unconditional reverse chain.

7. Teaching this note

About **35 minutes** at the board, then **~6 minutes** of video.

- **0–10 min.** MSE on ε . Same x_t formula as **12.1**.
 - **10–22 min.** Write one reverse-step mean. Circle the ε_θ term.
 - **22–33 min.** Worked scalar reverse step, then the Lab 12 oracle story.
 - **Then** play Umar Jamil **13:00–16:36** (training loop, then sampling loop). Pause when he draws one denoising step.
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8. Worked example

Reuse $x_0 = 1.0$, $\bar{\alpha}_t = 0.64$, $\varepsilon = 2.0$, so $x_t = 2.0$ as in note **12.1**. Take $\alpha_t = 0.64$ for a single-step cartoon (so $\sqrt{\alpha_t} = 0.8$, $1 - \alpha_t = 0.36$, $\sqrt{1 - \bar{\alpha}_t} = 0.6$).

Suppose $\varepsilon_\theta = \varepsilon = 2.0$ exactly, and drop $\sigma_t z$ for the mean:

$$x_{t-1} = \frac{1}{0.8} \left(2.0 - \frac{0.36}{0.6} \cdot 2.0 \right) = 1.25 (2.0 - 1.2) = 1.25 \times 0.8 = 1.0.$$

The reverse mean lands on x_0 . That is the oracle: if you predict the true noise, one algebraic step peels it off (here $\alpha_t = \bar{\alpha}_t$, a first-step special case). At a generic t you do not jump all the way to x_0 ; you move **closer**. Lab 12 checks $\|x_t - x_0\|$ vs $\|\hat{x} - x_0\|$ at $t = 20$ and wants the second smaller.

If $\varepsilon_\theta = 0$ (the net refuses to denoise), the subtracted term vanishes and you mostly rescale x_t , which is not a sample from the data.

 Oracle reverse mean: $\langle \text{math xmlns=$

2.0 peels back to 1.0" src="graphics/12.2-diffusion-reverse/reverse-mean.png" />

Sampling is an **iterative** procedure. You pay T network calls (or fewer with DDIM). A GAN still samples in one pass and fights collapse instead.

9. Where students get stuck

- Training a U-Net in Lab 12. The lab uses the **true** posterior mean because you know x_0 .
- Omitting t as an input to ε_θ . The same x_t is a different problem at $t = 10$ and at $t = 900$.
- Forgetting $\sigma_t z$ on the way down and then wondering why every sample looks identical.

10. Video

Watch [Umar Jamil, How diffusion models work](#), **13:00–16:36**.

Pause on the training objective (predict the noise) and on the sampling loop (many reverse steps). Same URL as note **12.1**; do not replay 7:35–13:00 unless needed.

11. Practice

1. If $\varepsilon_\theta = \varepsilon$ exactly, what should happen to $\|x_{t-1} - x_0\|$ relative to $\|x_t - x_0\|$, in expectation, for mid-range t ?
2. Why add $\sigma_t z$ on the way down instead of always taking the mean?
3. You may not train a score network in Lab 12. What replaces ε_θ for that one reverse step?

4. $x_t = 3$, $\alpha_t = 0.25$, $\tilde{\alpha}_t = 0.25$, $\varepsilon_\theta = 2$, no extra noise. Compute the reverse mean. ($\sqrt{\alpha_t} = 0.5$, $\sqrt{1 - \tilde{\alpha}_t} = \sqrt{0.75} \approx 0.866$, $1 - \alpha_t = 0.75$.)
5. Training samples t uniformly from $1 \dots T$ and one ε . Why is that enough, instead of a full reverse chain inside the loss?